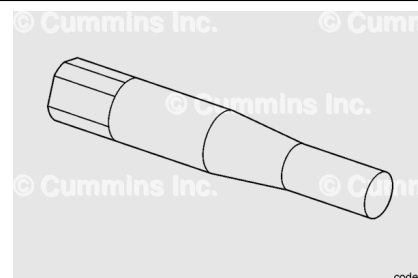


## Trainings ▾

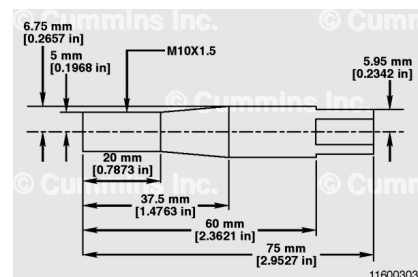
### Install

A manifold-centering dowel tool should be fabricated locally to assist in the installation of the gaskets and the exhaust manifold sections.



Use the dimensioned illustration to fabricate four manifold centering dowels from a length of 14 mm [0.5511 in] diameter standard steel bar. The tolerance for dimensions in the illustration is  $\pm 0.1$  mm [ $\pm 0.0039$  in].

This tool is intended for manual use.



### Industrial and Power Generation

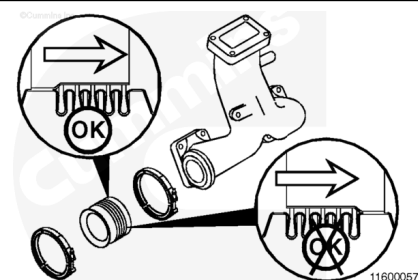
**Note :** The exhaust manifold for the QSK60 is shown. The procedures for the QSK45 are similar.

The exhaust manifold **must** be assembled as it is being installed.

Do **not** tighten the bellows clamps until all the exhaust manifolds are installed.

Apply anti-sieze compound, Part Number 3824879, to the exhaust mounting capscrews.

The exhaust manifold bellows are directional. Make sure they are installed in the proper manner when installing the exhaust manifold.



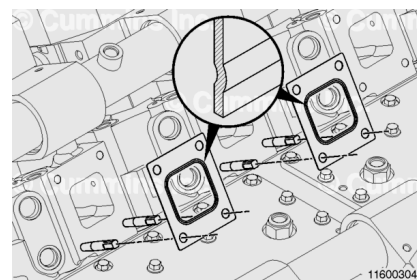
Ensure that the threads in the cylinder heads are free from any debris. If needed, the threads may be cleaned by tapping out the thread.

All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 5 and 6, or left bank cylinder heads number 3 and 4.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Note there are two different capscrews available for use; a stainless steel capscrew and a steel capscrew. The stainless steel capscrew can be identified as it will have the part number on the capscrew head. The stainless steel capscrew **must** also be fitted with the new spacers, these can be identified as they have two grooves around the circumference.

For the stainless steel capscrew, lubricate the threads and the capscrew underhead with Loctite® LB8009 anti-seize lubricant. Do **not** use copper grease to lubricate the stainless steel capscrew.

For the steel capscrew, lubricate the threads and capscrew underhead with copper grease.

Install capscrews and spacers into the upper holes and center the manifold section.

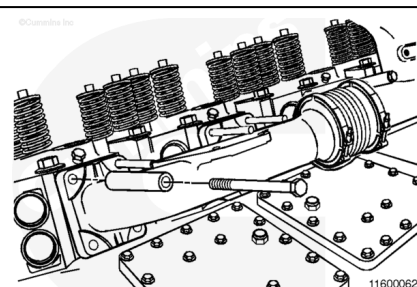
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]

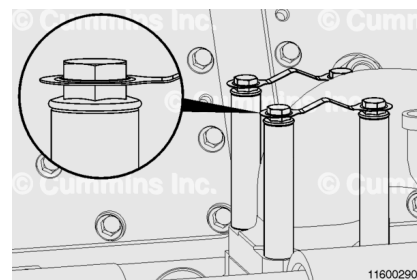


### **⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**

**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

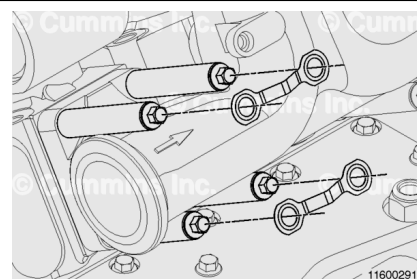
The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.



Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

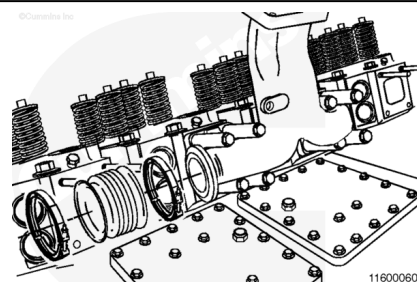
**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure the bellows are installed in the proper direction.

All of the bellows clamps **must** be installed with the clamp screw toward the opposite bank of the engine, with the screw pointing up.

Loosely install the bellows clamp and the bellows on the manifold section just installed.

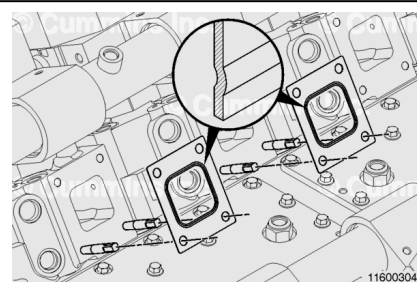


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 3 and 4, or left bank cylinder heads number 5 and 6.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install capscrews and spacers into the upper holes and center the manifold section.

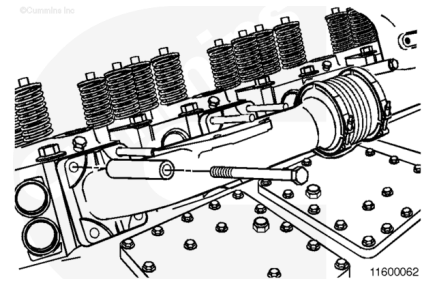
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

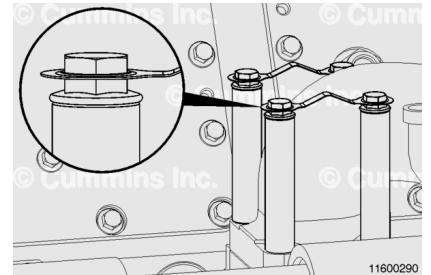
Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]



**⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**



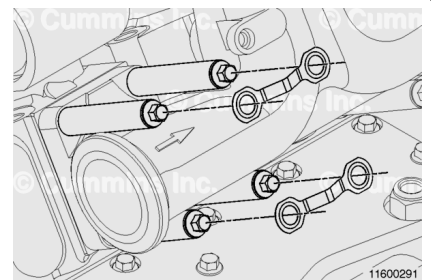
**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Install the bellows clamp.

Tighten the bellows clamps.

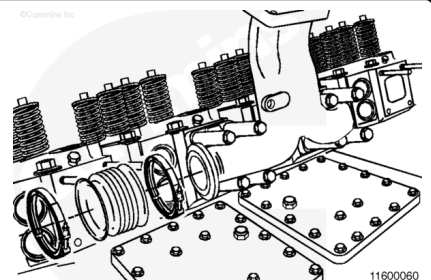
**Note :** There are two types of V-band clamps on the QSK45 and QSK60. They can be identified by the nut size.

**Torque Value:**

8mm (5/16 inch): 17 n•m [ 150 in-lb ]

**Torque Value:**

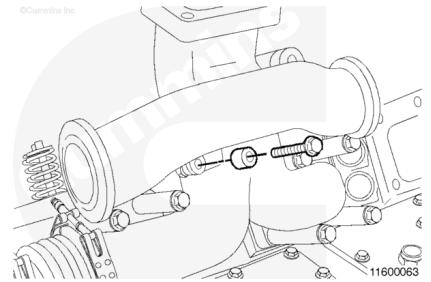
6mm (1/4 inch): 10 n•m [ 89 in-lb ]



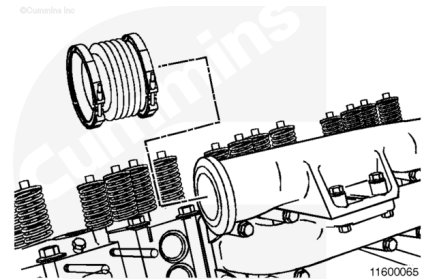
Ring dowels are used to locate and position the crossover section.

Position the crossover exhaust manifold section to the exhaust manifold section for right bank cylinders number 7 and 8, or left bank cylinders number 1 and 2.

Do **not** tighten the mounting capscrews.



Loosely install the bellows clamps and bellows to the ends of the upper exhaust manifold section.

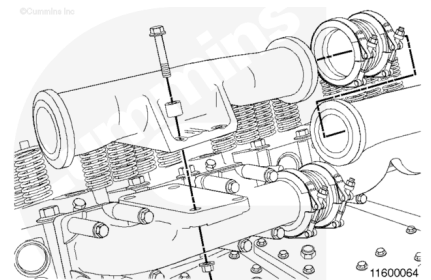


Position the crossover exhaust manifold section on top of the exhaust manifold section for right bank cylinders number 3 and 4, or left bank number 5 and 6.

Do **not** tighten the mounting capscrews.

Loosely install the bellows clamps and bellows to the upper exhaust manifold section mounted to the turbocharger mounting exhaust manifold section for right bank cylinders number 5 and 6, left bank cylinders number 3 and 4.

Do **not** tighten the bellows clamps.

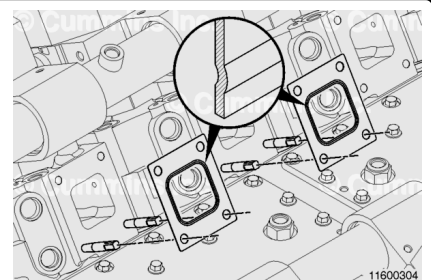


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 1 and 2, or left bank cylinder heads number 7 and 8.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

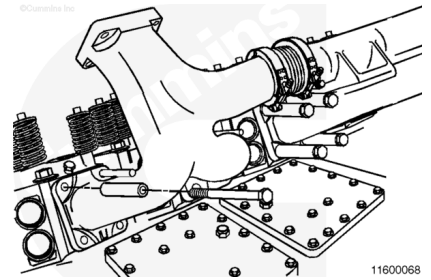
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

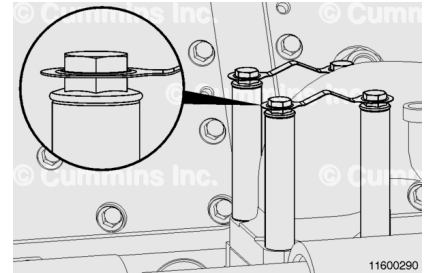
Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]



**⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**



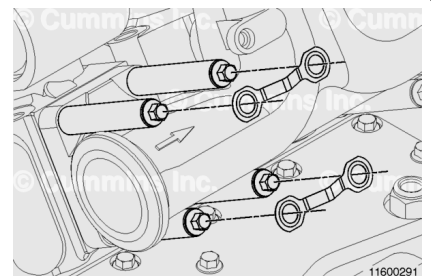
**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

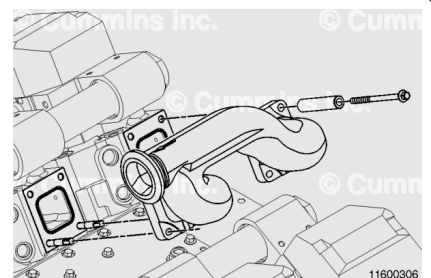
**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 7 and 8, or left bank cylinder heads number 1 and 2.



Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.

Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

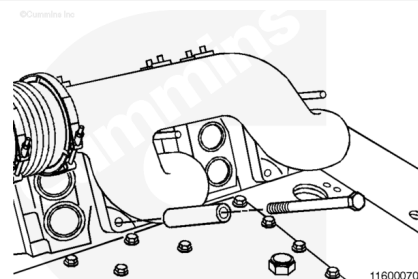
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

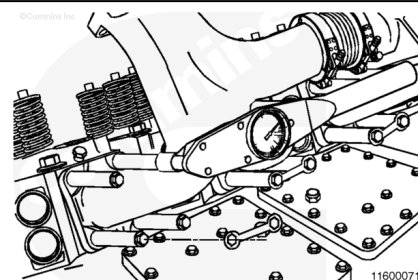
Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]



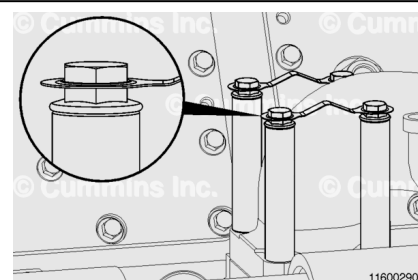
Tighten all exhaust manifold mounting capscrews in the exhaust manifold crossover sections that were **not** tightened earlier in the installation procedure.

**Torque Value:** 47 n•m [ 35 ft-lb ]



**⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**



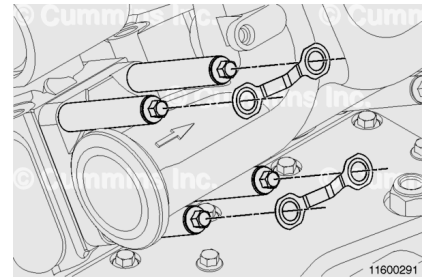
**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

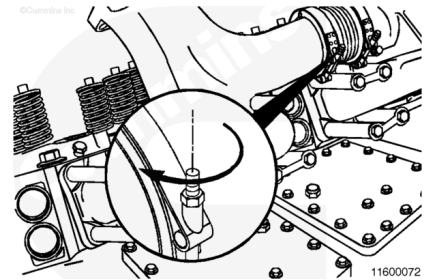
**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure all the bellows clamps are oriented with the clamp screw toward the opposite bank of the engine with the screw pointing up. This allows accessibility after complete assembly.

Tighten all the exhaust manifold bellows clamps.

**Note :** There are two types of V-band clamps on the QSK45 and QSK60. They can be identified by the nut size.



**Torque Value:**

8mm (5/16 inch): 17 n·m [ 150 in-lb ]

**Torque Value:**

6mm (1/4 inch): 10 n·m [ 89 in-lb ]

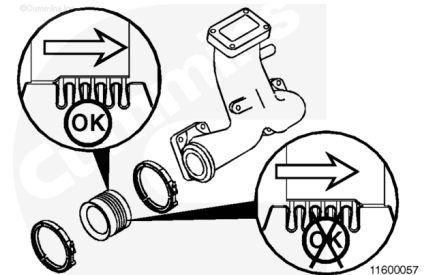
## QSK60 Marine Applications

The exhaust manifold **must** be assembled as it is being installed.

Do **not** tighten the bellows clamps until all the exhaust manifolds are installed.

Apply anti-sieze compound, Part Number 3824879, to the exhaust mounting capscrews.

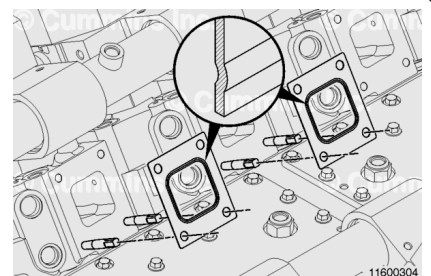
The exhaust manifold bellows are directional. Make sure they are installed in the proper manner when installing the exhaust manifold.



All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 5 and 6, or left bank cylinder heads number 3 and 4.





Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.

Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

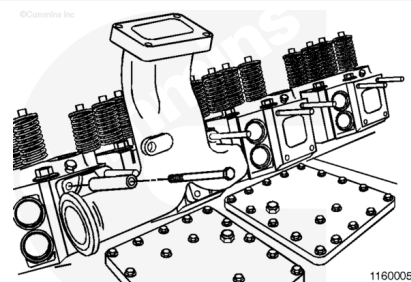
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

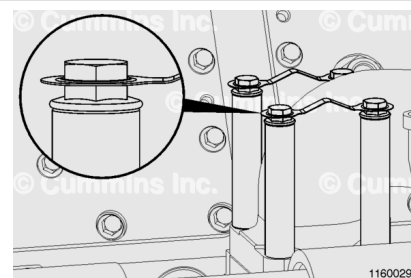
Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]



**⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**



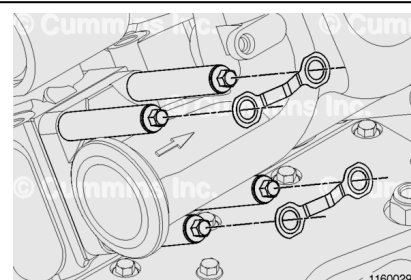
**Note :** When installing the lockplate over the capscrew head, install the lockplate approximately one half of the height. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.

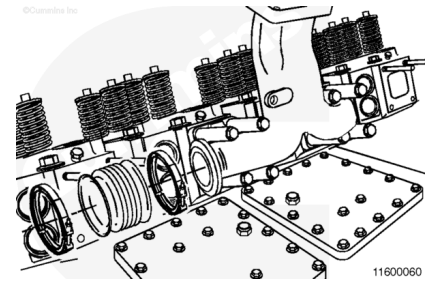


Install the bellows gasket onto each end of the bellows.

Install the bellows in the proper direction

All of the bellows clamps **must** be installed with the clamp screw toward the opposite bank of the engine, with the screw pointing up.

Loosely install the bellows clamp and bellows on the manifold section just installed.

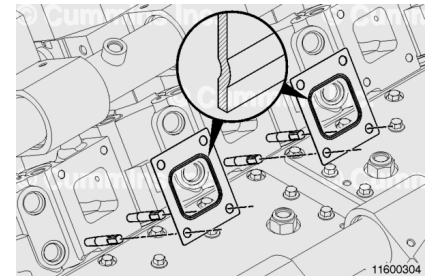


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 3 and 4, or left bank cylinder heads number 5 and 6.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

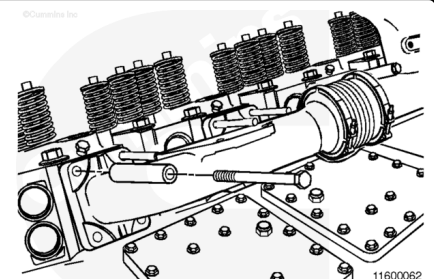
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

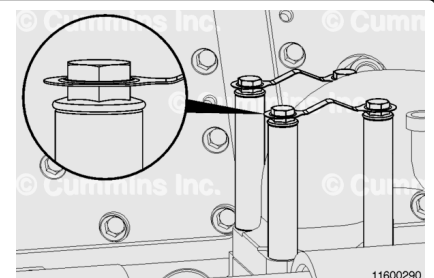
**Torque Value:** 47 n•m [ 35 ft-lb ]



### **⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**

**Note :** When installing the lockplate over the capscrew head, install the lockplate approximately one half of the height. Installing the lockplate until it bottoms can cause it to break.

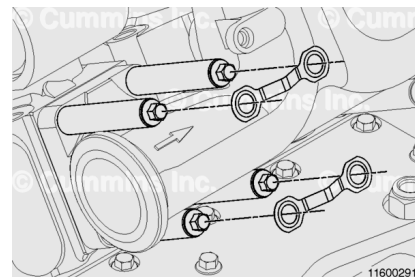


The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

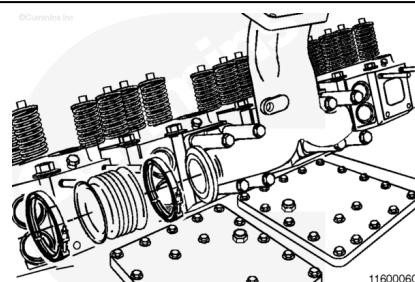
**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Install the bellows clamp.

Tighten the marine application bellows clamps.

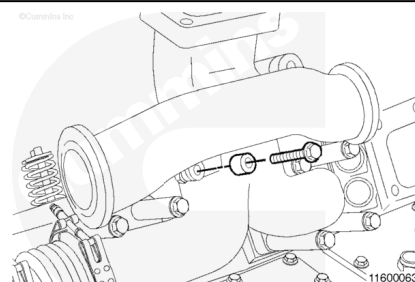
**Torque Value:** 17 n•m [150 in-lb]



Ring dowels are used to locate and position the crossover section.

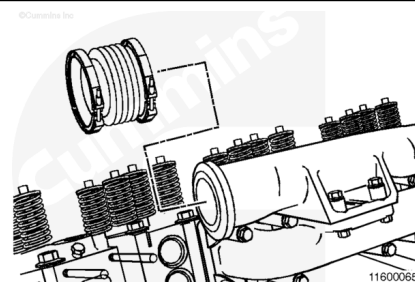
Position the crossover exhaust manifold section to the exhaust manifold section for right bank cylinders number 7 and 8, or left bank cylinders number 1 and 2.

Do **not** tighten the mounting capscrews.



Install the bellows gasket onto each end of the bellows.

Loosely install the bellows clamps and bellows to the ends of the upper exhaust manifold section.

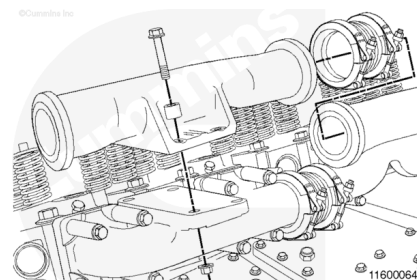


Position the crossover exhaust manifold section on top of the exhaust manifold section for right bank cylinders number 3 and 4, or left bank number 5 and 6.

Do **not** tighten the mounting capscrews.

Loosely install the bellows clamps and bellows to the upper exhaust manifold section mounted to the turbocharger mounting exhaust manifold section for right-bank cylinders number 5 and 6, left-bank cylinders number 3 and 4.

Do **not** tighten the bellows clamps.

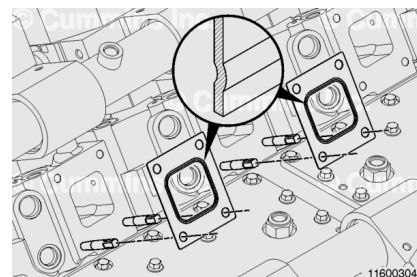


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 1 and 2, or left bank cylinder heads number 7 and 8.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

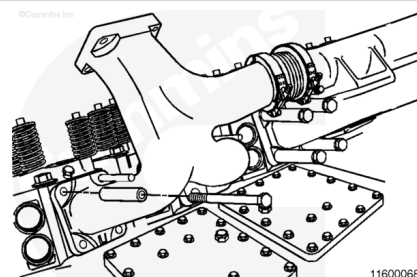
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

Tighten all eight capscrews in the exhaust manifold section.

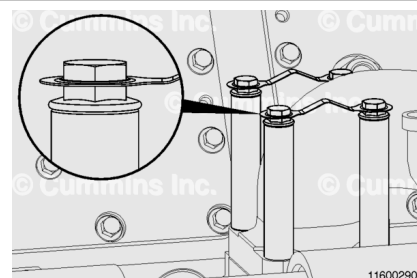
**Torque Value:** 47 n•m [ 35 ft-lb ]



### **⚠ CAUTION ⚠**

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**

**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

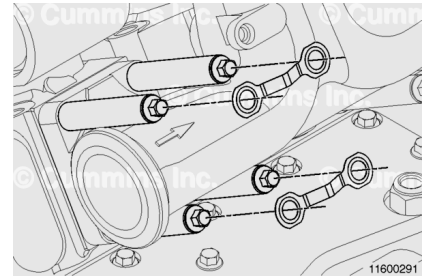


The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.

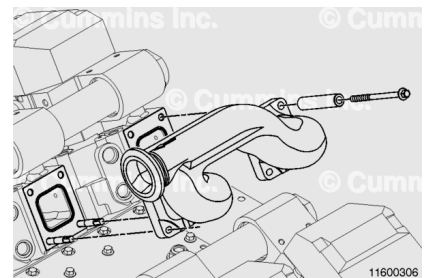


All exhaust manifold gaskets **must** be installed with the bead toward the manifold.

Two manifold centering dowels **must** be used to position each exhaust manifold gasket onto each exhaust port.

Hand tighten the centering dowels into the lower holes of right bank cylinder heads number 7 and 8, or left bank cylinder heads number 1 and 2.

Install the exhaust manifold gaskets onto the dowels. Make sure the gasket side with the embossed bead (marked "OUT") is facing away from the cylinder head exhaust port. Make sure the gasket is properly aligned.



Place the exhaust manifold section onto the centering dowels.

Install the capscrews and spacers into the upper holes and center the manifold section.

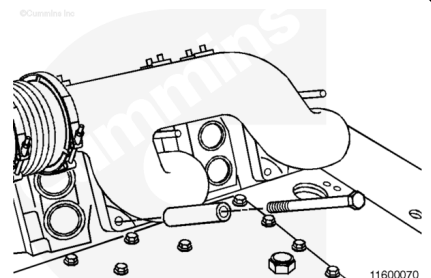
Tighten the upper capscrews.

**Torque Value:** 10 n•m [ 89 in-lb ]

Remove the centering dowels from the lower holes and install the capscrews and spacers. Hand tighten the capscrews.

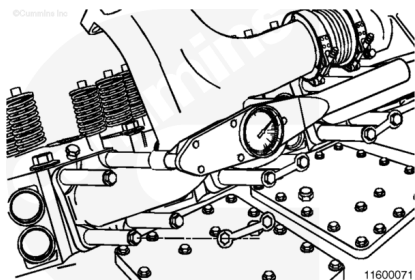
Tighten all eight capscrews in the exhaust manifold section.

**Torque Value:** 47 n•m [ 35 ft-lb ]



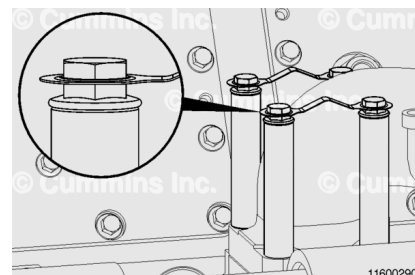
Tighten all exhaust manifold mounting capscrews in the exhaust manifold crossover sections that were **not** tightened earlier in the installation procedure.

**Torque Value:** 47 n•m [ 35 ft-lb ]



### ⚠ CAUTION ⚠

**Do not install the lockplate upside down. The lockplate will either break or not be retained properly on the capscrew head.**



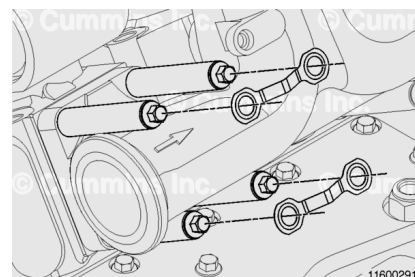
**Note :** Install the lockplate approximately one half of the height of the capscrew head. Installing the lockplate until it bottoms can cause it to break.

The illustration shows the correct orientation of the lockplate over the capscrew. Note the raised edge faces away from the capscrew head flange.

Install the exhaust manifold mounting capscrew lockplates. Make sure the lockplate V-bend is facing away from the cylinder head.

Use service tool, Part Number 4919376, for correct installation depth.

**Note :** For ease of installation, complete installation of one bank of exhaust manifolds and lockplates prior to installing the second bank.



Make sure all the bellows clamps are oriented with the clamp screw toward the opposite bank of the engine with the screw pointing up. This allows accessibility after complete assembly.

Tighten all the marine application bellows clamps.

**Torque Value:** 17 n•m [150 in-lb]

